

Number:



Roseville College

Year 12

Trial Higher School Certificate Examination

2001

EXTENSION 1 MATHEMATICS

Time Allowed: 2 hours, plus 5 minutes reading time.

Instructions

- All questions are of equal value.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Start each question on a new page. Write your number on each page.
- Staple each question separately

QUESTION 1. Start a new page (12 marks)

- (a) Use the substitution $u = x^2 + 2$ to evaluate

$$\int_0^1 \frac{x}{x^2 + 2} dx \quad (3)$$

- b) Solve for x if $\frac{4}{x-2} > 3$ (3)

- (c) Find the exact value of $\tan\left(2 \tan^{-1} \frac{3}{4}\right)$ (2)

- (d) A box contains 12 jellybeans of which 5 are red, 4 are blue and 3 are white. If 3 jellybeans are picked up at once what is the probability that all three are different colours? (2)

- (e) Sketch a continuous smooth curve which satisfies the following conditions

$$\begin{aligned} f(0) &= 1 \\ f'(x) &< 0 \text{ and } f''(x) > 0 \text{ for } 0 < x < 2 \\ f'(2) &= 0 \\ f(2) &= -2 \\ f''(x) &< 0 \text{ and } f''(x) < 0 \text{ for } x > 2 \end{aligned} \quad (2)$$

QUESTION 2. Start a new page (12 marks)

- (a) State the domain and range

$$f(x) = 4 \sin^{-1}\left(\frac{x}{3}\right) \quad (3)$$

- (b) (i) Show that the equation $x^3 + x - 3 = 0$ has 1 root between 1.2 and 1.3

- (ii) Taking 1.2 as the first approximation to the root, use Newton's method once to find a second approximation. (3)

- (c) A polynomial $P(x)$ of degree three, has zeros at $x = -2$, $x = -1$ and $x = 1$ and a remainder of 36 when divided by $(x - 2)$. Find $P(x)$, expressing it in the form

$$p_0 x^3 + p_1 x^2 + p_2 x + p_3 \quad (3)$$

- (d) The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ meets the directrix at K

- (i) Show that the coordinates of K are $\left(\frac{ap^2 - a}{p}, -a\right)$ (1)

- (ii) Prove that angle PSK is a right angle, where S is the focus (2)

QUESTION 3. Start a new page (12 marks)

(a) The acceleration of a particle is given by $4(1+x)$, where x is the displacement from the origin.

If initially, the particle is at the origin with a velocity of 2ms^{-1} ,

(i) show that $v = 2(x+1)$ (2)

(ii) show that $x = e^{2t} - 1$ (2)

(iii) find its acceleration after 1 second (2)

(b) Express the solution to the equation

$\sin 2\theta = \sin \theta$ in general form, θ in radians (2)

(c) Find

(i) $\int \frac{dx}{\sqrt{9-4x^2}}$ (2)

(ii) $\int \sin^2 x dx$ (2)

QUESTION 4. Start a new page (12 marks)

(a) Show that

$$\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{2} \quad (2)$$

(b) Kool has decided to invest in a superannuation fund. She calculates that she will need \$1 000 000 if she is to retire in 20 years time and maintain her present lifestyle. The superannuation fund pays 12% per annum interest on her investments.

(i) Kool invests \$P at the beginning of each year. Show that at the end of the first year her investment is worth $\$P(1.12)$ (1)

(ii) Show that at the end of the third year the value of her investment is given by the expression $\$P(1.12)(1.12^2 + 1.12 + 1)$ (2)

(iii) Find a similar expression for the value of her investment after 20 years and hence calculate the value of P needed to realise the total of \$1 000 000 required for his retirement. (3)

(c) The daily growth of the population of a colony of insects is 10% of the excess of the population over 1.2×10^6 . At $t = 0$ the population is 2.7×10^6 (Given $P = N + Ae^{kt}$)

(i) Determine the population after $3\frac{1}{2}$ days. (2)

(ii) If a scientist checks the population each day, which is the first day on which she should notice the original population has tripled? (3)

QUESTION 5. Start a new page (12 marks)

- (a) A sphere is being heated so that its surface area is increasing at a constant rate of 15mm^2 per second. Find the rate of increase of the volume when the radius is 5mm . (3)

- (b) Find the value of the constant m if e^{mx} satisfies the differential equation

$$\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6y = 0 \quad (3)$$

- (c) A javelin is thrown across level ground from a height of 2m at a speed of 20m/s at an angle of 60° to the horizontal. Taking acceleration due to gravity as 10m/s^2 , find

- (i) the height reached (2)
- (ii) the time the javelin is in the air (2)
- (iii) the length of the throw (2)

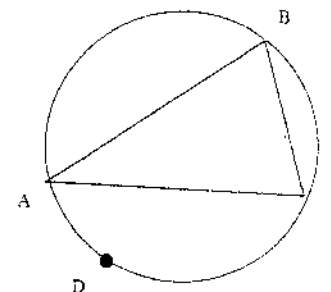
QUESTION 6. Start a new page (12 marks)

- (a) A particle moves along a straight line with a velocity given by $\frac{1}{2}v^2 = 18 - 2x^2$, where x is the distance from a fixed point O on the line.

- (i) show that the motion is simple harmonic (1)

- (ii) find the period and amplitude of the motion of the motion (2)

(b)



ABCD are four points on a circle centre O and radius R units, such that BD is a diameter. A, B, C are joined to form a triangle in which $AB=c$ units, $BC=a$ units and $AC=b$ units. Show, giving reasons, that

(i) $\sin \angle BAC = \frac{a}{2R}$ (3)

(ii) $\text{Area } \triangle ABC = \frac{abc}{4R}$ (3)

- (c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $A \sin(x + \alpha)$ (2)

- (ii) Use this to solve $\sin x + \sqrt{3} \cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$ (2)

QUESTION 7. Start a new page (12 marks)

(a) Prove that for all positive integers n , $9^{n+2} - 4^n$ is divisible by 5. (4)

(b) Evaluate

$$\int_0^{\frac{1}{2}} \frac{dx}{1+4x^2} \quad (3)$$

(c) The line $y = 2x + 2$ cuts the line segment AB at some point C. If A is the point $(-2,3)$ and B is the point $(4,3)$ find the ratio of AC:CB. (2)

(d) If $y = \frac{1}{2} \cdot (e^x - e^{-x})$, show that $x = \log_e (y + \sqrt{y^2 + 1})$ (3)

END OF PAPER

Question 1

(a) $\int_0^1 \frac{x}{x^2+2} dx$

$= \frac{1}{2} \int_0^1 \frac{1}{u} du$

$= \frac{1}{2} [\ln u]_0^1$

$= \frac{1}{2} \ln(3)$

(b) $4(x-2) > 3(x-2)^2$
 $4x-8 > 3x^2-12x+12$
 $0 > 3x^2-16x+20$
 $3x^2-16x+20 < 0$
 $(3x-10)(x-2) < 0$
 $2 < x < 3\frac{1}{3}$

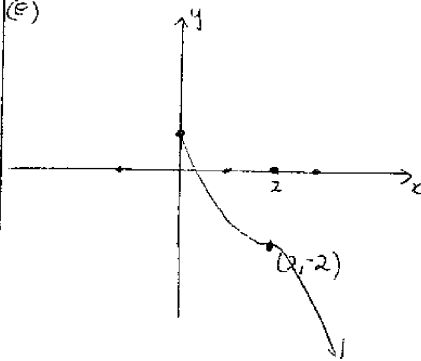
(c) $\tan(2 \tan^{-1} \frac{3}{4})$

Let $\theta = \tan^{-1} \frac{3}{4}$
 $\therefore \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$
 $= \frac{2 \times \frac{3}{4}}{1 - \frac{9}{16}}$
 $= \frac{24}{7}$

(d) SR, 4B, 3W

$P(R, B, W) \text{ or } (BRW) \text{ or } (R, W, B)$
 $= \left(\frac{5}{12} \times \frac{4}{11} \times \frac{3}{10} \right) \times 6$

(e)



Question 2

(a) $f(x) = 4 \sin^{-1} \left(\frac{x}{3} \right)$

domain $-1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$

range $-2\pi \leq y \leq 2\pi$

(b) (i) $F(x) = x^3 + x - 3$

$F(1.2) = -0.072 < 0$

$F(1.3) = 0.497 > 0$

$F(1.2) < 0$ $F(1.3) > 0$ root between

(ii) $F(x) = 1.2 - \frac{F(1.2)}{F'(1.2)}$
 $= 1.2 - \frac{(-0.072)}{3(1.2)^2 + 1}$
 $= 1.213533835$

(c) $P(x) = K(x+2)(x+1)(x-1) = 0$

Now $P(2) = 36$

$\therefore 36 = K(4)(3)(1)$

$\therefore K = 3$

$\therefore P(x) = 3(x+2)(x+1)(x-1)$

$= (3x+6)(x^2-1)$

$= 3x^3 - 3x + 6x^2 - 6$

$P(x) = 3x^3 + 6x^2 - 3x - 6$

(d) $A + P(2ap, ap^2) \quad m = p$

(i) $y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$

K is where $y = -a$

$-a = px - ap^2$

$px = ap^2 - a$

$x = \frac{ap^2 - a}{p}$

$\therefore K \text{ is } \left(\frac{ap^2 - a}{p}, -a \right)$

(ii) $m_{PS} = \frac{ap^2 - a}{2ap} = \frac{p^2 - 1}{2p}$

$m_{SK} = \frac{a + a}{-ap^2 + a} = \frac{-2ap}{ap^2 - a}$

$= \frac{2ap}{a - ap^2}$

$= \frac{2p}{1 - p^2}$

Since $m_{PS} \times m_{SK} = -1$

$\angle PSK \text{ is } 90^\circ$

Question 3

(a) $\ddot{x} = 4(1+x)$

$\frac{d^2v}{dx^2} = 4(1+x)$

(i) $\frac{1}{2}v^2 = 4x + 2x^2 + C$

when $v = 0$, $x = 0$

$2 = C$

$\therefore \frac{1}{2}v^2 = 4x + 2x^2 + 2$

$v^2 = 8x + 4x^2 + 4$

$v = \sqrt{4x^2 + 8x + 4}$

$v = 2\sqrt{x^2 + 2x + 1}$

$v = 2\sqrt{(x+1)^2}$

$v = 2(x+1)$

(ii) $x = e^{2t} - 1$

Now $\frac{dx}{dt} = 2(x+1)$

$\frac{dt}{dx} = \frac{1}{2(x+1)}$

$t = \frac{1}{2} \int \frac{1}{x+1} dx$

$t = \frac{1}{2} \ln(x+1) + K$

when $t = 0$, $x = 0$, $\therefore K = 0$

$2t = \ln(x+1)$

$e^{2t} = x+1$

$\therefore x = e^{2t} - 1$

(ii) $\ddot{x} = 2e^{2t}$

$\ddot{x} = 4e^{2t}$

when $t = 1$

$\ddot{x} = 4e^2 \text{ m/s}^2$

$$b) \sin 2\theta = \sin \theta$$

$$2 \sin \theta \cos \theta = \sin \theta$$

$$2 \sin \theta \cos \theta - \sin \theta = 0$$

$$\sin \theta (2 \cos \theta - 1) = 0$$

$$\sin \theta = 0, \cos \theta = \frac{1}{2}$$

$$\theta = (-1)^n \sin^{-1} 0, 2\pi n + \cos^{-1} \frac{1}{2}$$

$$= n\pi + (-1)^n \cdot 0, 2\pi n \pm \frac{\pi}{3}$$

$$= n\pi, 2\pi n \pm \frac{\pi}{3}$$

$$c) (i) \int \frac{dx}{\sqrt{9-4x^2}}$$

$$= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$$

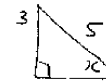
$$(ii) \int \sin^3 x dx$$

$$= \int \frac{1}{2} - \frac{1}{2} \cos 2x dx$$

$$= \frac{x}{2} - \frac{1}{4} \sin 2x + C$$

Question 4

$$a) \cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{2}$$

$$\text{let } x = \cos^{-1} \frac{4}{5}$$


$$y = \cos^{-1} \frac{3}{5}$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$= \frac{4}{5} \times \frac{3}{5} - \frac{3}{5} \times \frac{4}{5}$$

$$= 0$$

$$\therefore x+y = \cos^{-1}(0)$$

$$x+y = \frac{\pi}{2}$$

$$\therefore \cos^{-1} \frac{4}{5} + \cos^{-1} \frac{3}{5} = \frac{\pi}{2}$$

$$(b) (i) P(1+r)^n = P(1.12)^1$$

$$(ii) B_2 = P(1.12)^2 + P(1.12)$$

$$B_3 = P(1.12)^3 + P(1.12)^2 + P(1.12)$$

$$= P(1.12)(1.12^2 + 1.12 + 1)$$

$$(ii) B_{20} = P(1.12)(1.12^{19} + 1.12^{18} + \dots + 1)$$

$$1000000 = P(1.12) \left(\frac{1.12^{20} - 1}{0.12} \right)$$

$$1000000 = P(1.12) \left(\frac{1.12^{20} - 1}{0.12} \right)$$

$$120000 = P(1.12)(1.12^{20} - 1)$$

$$P = \frac{120000}{(1.12)(1.12^{20} - 1)}$$

$$= \$12391.77$$

$$(c) (i) P = N + Ae^{0.1t}$$

$$\text{at } t=0, P = 2.7 \times 10^6$$

$$2.7 \times 10^6 = 1.2 \times 10^6 + A$$

$$A = 1.5 \times 10^6$$

$$\therefore P = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.1t}$$

$$\text{When } t=3.5$$

$$P = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.35}$$

$$= 3328601.323$$

$$= 3.3 \times 10^6$$

$$\text{when } P = 8.1 \times 10^6$$

$$8.1 \times 10^6 = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.1t}$$

$$6.9 \times 10^6 = 1.5 \times 10^6 e^{0.1t}$$

$$4.6 = e^{0.1t}$$

$$\ln(4.6) = 0.1t$$

$$t = 15.26$$

$$\therefore \text{On } 16^{\text{th}} \text{ day}$$

Question 5

$$(a) \frac{dA}{dt} = 15$$

$$\text{at}$$

$$\frac{dv}{dt} = \frac{dv}{dr} \times \frac{dr}{dt}$$

$$= 4\pi r^2 \times \frac{dr}{dt}$$

$$V = \frac{4\pi r^3}{3}$$

$$\frac{dv}{dr} = 4\pi r^2$$

$$\text{Now } \frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt}$$

$$= \frac{1 \times 15}{8\pi r} \cdot A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\therefore \frac{dv}{dt} = 4\pi r^2 \times \frac{15}{28\pi r}$$

$$= \frac{15r}{2}$$

$$\text{When } r=5$$

$$\frac{dv}{dt} = 37\frac{1}{2} \text{ mm}^3/\text{s}$$

$$\text{at}$$

$$(b) y = e^{mx}$$

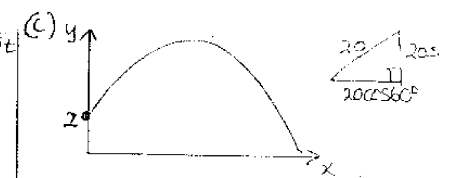
$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$$

$$m^2 e^{mx} - m e^{mx} - 6 e^{mx} = 0$$

$$e^{mx} (m^2 - m - 6) = 0$$

$$e^{mx} (m-3)(m+2) = 0$$

$$m = 2, 3$$



$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{x} = 10$$

$$\dot{y} = -10t + 10\sqrt{3}$$

$$x = 10t$$

$$y = -5t^2 + 10\sqrt{3}t + 2$$

$$(i) \text{ height reached } \rightarrow \dot{y} = 0$$

$$-10t + 10\sqrt{3} = 0$$

$$t = \sqrt{3} \text{ s}$$

$$\therefore y = -5(3) + 10(3) + 2$$

$$= -15 + 30 + 2$$

$$y = 17 \text{ m}$$

$$(ii) \text{ time of flight } y = 0$$

$$-5t^2 + 10\sqrt{3}t + 2 = 0$$

$$5t^2 - 10\sqrt{3}t - 2 = 0$$

$$t = \frac{10\sqrt{3} \pm \sqrt{300 + 40}}{10}$$

$$t = \sqrt{3}, \sqrt{340}$$

$$t = 31.937 \text{ s}$$

$$(iii) \text{ when } t = \sqrt{1020}$$

$$x = 10\sqrt{1020} \text{ m } (319.37)$$

Question 6

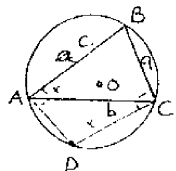
$$(a) \frac{1}{2} v^2 = 18 - 2x^2$$

$$(i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4x \text{ S.H.M.}$$

$$(ii) T = \frac{2\pi}{\omega} = \pi$$

$$\text{When } v=0 \quad 18 - 2x^2 = 0$$

$$\therefore a=3 \quad x=3$$



$\angle BCD = 90^\circ$ ($\angle$ in Semi-circle)

$\angle BDC = \angle BAC$ ($\angle$'s standing on same arc)

$$\therefore \sin \angle BDC = \frac{a}{R}$$

$$\sin \angle BDC = \frac{a}{R}$$

$$\therefore \sin \angle BAC = \frac{a}{R}$$

$$\begin{aligned} \text{(ii) Area } \triangle ABC &= \frac{1}{2} \cdot C \cdot b \cdot \sin \angle C \\ &= \frac{1}{2} \cdot C \cdot b \cdot \frac{a}{R} \\ &= \frac{abc}{4R} \end{aligned}$$

$$\begin{aligned} \text{(c) (i) } \sin x + \sqrt{3} \cos x &= A \sin(x + \alpha) \\ A &= \sqrt{1+3} = 2 \\ \alpha &= \tan^{-1} \sqrt{3} = \frac{\pi}{3} \end{aligned}$$

$$\therefore \sin x + \sqrt{3} \cos x = 2 \sin(x + \frac{\pi}{3})$$

$$\begin{aligned} \text{(ii) } \sin x + \sqrt{3} \cos x &= 2\sqrt{3} \\ \therefore 2 \sin(x + \frac{\pi}{3}) &= 2\sqrt{3} \end{aligned}$$

$$\sin(x + \frac{\pi}{3}) = \sqrt{3}$$

$$\begin{aligned} x + \frac{\pi}{3} &= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3} \\ x &= 0, \pi/3, 2\pi \end{aligned}$$

Question 7

(a) Induction.

Step ① Assume true $n=k$

$$9^{k+2} - 4^k = 5M \quad M \text{ is int}$$

Step ② Prove true for $n=k+1$
R.T.P.

$$9^{k+3} - 4^{k+1} \text{ is divisible by 5}$$

Step ③ Proof

$$9^{k+3} - 4^{k+1} = 9 \cdot 9^{k+2} - 4^{k+1}$$

$$= 9(5M + 4^k) - 4^{k+1}$$

$$= 45M + 9 \cdot 4^k - 4 \cdot 4^k$$

$$= 45M + 5 \cdot 4^k$$

$$= 5(9M + 4^k)$$

which is divisible by 5

Step ④

Hence statement is true for $n=k+1$ when it is true for $n=k$.

Step ⑤

For $n=1$

$$9^2 - 4 = 80$$

$\therefore$ True for $n=1$

Step ⑥

Since true for $n=1$ by step ④ it will be true for $n=2$ and then $n=3$ and so on for all integers.

$$(b) \int_0^{\frac{1}{2}} \frac{dx}{1+4x^2} = \int_0^{\frac{1}{2}} \frac{dx}{4(\frac{1}{4}+x^2)}$$

$$= \frac{1}{4} \left[\tan^{-1} 2x \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{4} \left[\tan^{-1}(1) - \tan^{-1}(0) \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{4} \right]$$

$$= \frac{\pi}{16}$$

$$(c) A(-2, 3) \quad B(4, 3)$$

$$m_{AB} = 0$$

Equation $y=3$.

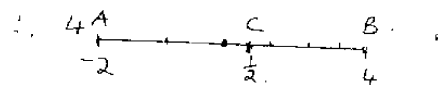
sub into $y=2x+2$

$$3=2x+2$$

$$2x=1$$

$$x = \frac{1}{2}$$

$\therefore C$ is $(\frac{1}{2}, 3) \rightarrow (m, n)$



$$\therefore AC = CB$$

$$2\frac{1}{2} : 3\frac{1}{2}$$

$$5 : 7$$

$$(d) y = \frac{1}{2} (e^x - e^{-x})$$

$$2y = e^x - e^{-x}$$

$$0 = e^{2x} - 1 - 2y \cdot e^x$$

$$0 = e^{2x} - 2y e^x - 1$$

$$e^x = \frac{2y \pm \sqrt{4y^2 + 4}}{2}$$

$$e^x = \frac{2y \pm 2\sqrt{y^2 + 1}}{2}$$

$$e^x = y \pm \sqrt{y^2 + 1}$$

$$e^x > 0 \quad \therefore \sqrt{y^2 + 1} > 0$$

$$e^x = y + \sqrt{y^2 + 1}$$

$$x = \ln(y + \sqrt{y^2 + 1})$$